Shawnak Samaddar

US Citizen | 404-387-2296 | shawnaksa150@gmail.com | linkedin.com/in/shawnaksa | github.com/shawnaks

Education

Georgia Institute of Technology — Atlanta, Georgia

Bachelor's of Science in Computer Science, Minor in Mathematics — *Expected Graduation: December 2027*

- **Concentrations:** Artificial Intelligence and Information Internetworks
- **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Database Systems, Systems and Networks, Object Oriented Programming, Probability and Statistics, Combinatorial Analysis, Objects and Design

Experience

Data Engineering Intern — May 2026 – August 2026

Delta Dental Insurance — *Atlanta, GA*

- Engineered a Python and Snowflake ETL data pipeline for the National Provider File, distilling 53M+ raw records into 27M refined target rows ready for dashboarding
- Developed a Python based AI agent connected to Snowflake and Matillion to automate the creation and QA of complex data pipelines from business logic documents, saving 80% developer time on manual ETL tasks
- Led data validation at scale across numerous environments, ensuring consistent pipeline behavior across 600M+ records and 50 distinct data sources and investigating sources of discrepancies

Undergraduate Research Assistant - Safe Robotics Lab — December 2025 – Present

Georgia Institute of Technology — *Atlanta, GA*

- Teleoperated OpenArm bimanual robot using Python scripts in MuJoCo to execute real-time object manipulation
- Processed data from human robot operators to train robot learning model for autonomous actions

Data Analyst Intern — June 2024 – September 2025

NASA Genelab Multi-Omics AWG — *Remote, US*

- Processed and analyzed stained microscopy data corresponding to encoding genes to identify expression patterns
- Clustered biological samples using K-Means to uncover structural relationships across gene datasets
- Applied multiple machine learning models (Random Forest, Logistic Regression, Single-Layer Perceptron, SVM, Ridge, and Lasso) and combined results through union/intersection analysis for improved predictive reliability
- Communicated model insights and technical findings to the biological research team for result interpretation

Projects

RxGuard | *RAG, Pandas, MatPlotLib, React, Gemini* | https://devpost.com/software/guess-xp546j — February 2026

- Won 2nd place in Best Use of Actian VectorAI DB at Hacklytics GT hackathon out of 50+ teams
- Built semantic drug interaction search engine, surfacing more relevant patient cases than keyword queries
- Designed a 4 stage ELT pipeline batch processing data in batches of 20k records from the openFDA API
- Embedded record narratives into 384-dim vectors with all-MiniLM-L6-v2 improving retrieval matches by $\sim 30\%$
- Architected a Actian VectorAI DB deployment for local HIPAA-compliant semantic search

SnapFind | *Python, PyTorch, Numpy, Pandas, FastAPI, React, Typescript, Git* — September 2025

- Developed CNN-powered lost-and-found app to automate item tagging and streamline retrieval for students
- Spearheaded a team at HackGT, deciding the tech stack and managing PRs, workflows, and delivery milestones
- Combined and cleaned 12K+ images from 10 datasets by building preprocessing to enhance model robustness
- Achieved 92% classification accuracy by fine-tuning ResNet18 with dropout-regularized classification head
- Deployed PyTorch model via FastAPI on Render, integrating with React frontend for real-time image classification

Flight Fare Optimizer | *Python, NumPy, Pandas, Matplotlib, Google Colab, Jupyter* — November 2024

- Processed 250,000+ flight fare and route records with NumPy and Pandas to construct a weighted airport graph
- Benchmarked algorithm performance, analyzing runtime efficiency and execution time to inform design decisions
- Visualized airport network and algorithm performance using Matplotlib to determine optimal fare and algorithm

Technical Skills

Languages: Python, Java, C/C++, SQL, ASM, JavaScript
Data, AI & ML: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Snowflake, Matillion
Web & Cloud: AWS, React, Node.js, Next.js, FastAPI, REST API, Docker, Selenium
Developer Tools: Git, Claude Code, Github Copilot, VS Code, Visual Studio, PyCharm, IntelliJ, Jupyter
Addtl. Skills: Agile/Scrum, Project Management, Prompt Engineering, Figma, Webscraping